Manage LDAP server configuration

ONTAP 9.7 REST API reference

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Manage LDAP server configuration

Security authentication cluster LDAP endpoint overview

Overview

LDAP servers are used to centrally maintain user information. LDAP configurations must be set up to look up information stored in the LDAP directory on the external LDAP servers. This API is used to retrieve and manage cluster LDAP server configurations.

Examples

Retrieving the cluster LDAP information

The cluster LDAP GET request retrieves the LDAP configuration of the cluster.

The following example shows how a GET request is used to retrieve the cluster LDAP information:

```
# The API:
/api/security/authentication/cluster/ldap

# The call:
curl -X GET "https://<mgmt-ip>/api/security/authentication/cluster/ldap"
-H "accept: application/hal+json"

# The response:
{
"servers": [
  "10.10.10.10",
  "domainB.example.com"
],
"schema": "ad_idmu",
"port": 389,
"min_bind_level": "anonymous",
"bind_dn": "cn=Administrators,cn=users,dc=domainA,dc=example,dc=com",
"base_dn": "dc=domainA,dc=example,dc=com",
"base_scope": "subtree",
"use_start_tls": true,
"session_security": "none",
"_links": {
  "self": {
    "href": "/api/security/authentication/cluster/ldap"
  }
}
}
```

Creating the cluster LDAP configuration

The cluster LDAP POST operation creates an LDAP configuration for the cluster.

The following example shows how to issue a POST request with all of the fields specified:

```
# The API:
/api/security/authentication/cluster/ldap

# The call:
curl -X POST "https://<mgmt-ip>/api/security/authentication/cluster/ldap"
-H "accept: application/hal+json" -H "Content-Type: application/json" -d
"{ \"servers\": [ \"10.10.10.10\"\, \"domainB.example.com\" ], \"schema\":
\"ad_idmu\", \"port\": 389, \"min_bind_level\": \"anonymous\",
\"bind_dn\": \"cn=Administrators,cn=users,dc=domainA,dc=example,dc=com\",
\"bind_password\": \"abc\", \"base_dn\": \"dc=domainA,dc=example,dc=com\",
\"base_scope\": \"subtree\", \"use_start_tls\": false,
\"session_security\": \"none\"}"
```

The following example shows how to issue a POST request with a number of optional fields not specified:

```
# The API:
/api/security/authentication/cluster/ldap

# The call:
curl -X POST "https://<mgmt-ip>/api/security/authentication/cluster/ldap"
-H "accept: application/hal+json" -H "Content-Type: application/json" -d
"{ \"port\": 389, \"bind_dn\":
\"cn=Administrators,cn=users,dc=domainA,dc=example,dc=com\",
\"bind_password\": \"abc\", \"base_dn\": \"dc=domainA,dc=example,dc=com\",
\"session_security\": \"none\"}"
```

Updating the cluster LDAP configuration

The cluster LDAP PATCH request updates the LDAP configuration of the cluster.

The following example shows how a PATCH request is used to update the cluster LDAP configuration:

```
# The API:
/api/security/authentication/cluster/ldap

# The call:
curl -X PATCH "https://<mgmt-ip>/api/security/authentication/cluster/ldap"
-H "accept: application/json" -H "Content-Type: application/json" -d "{
\"servers\": [ \"55.55.55.55\" ], \"schema\": \"ad_idmu\", \"port\": 636,
\"use_start_tls\": false }"
```

Deleting the cluster LDAP configuration

The cluster LDAP DELETE request deletes the LDAP configuration of the cluster.

The following example shows how a DELETE request is used to delete the cluster LDAP configuration:

```
# The API:
/api/security/authentication/cluster/ldap

# The call:
curl -X DELETE "https://<mgmt-
ip>/api/security/authentication/cluster/ldap" -H "accept:
application/hal+json"
```

Delete the LDAP configuration for the cluster

DELETE `/security/authentication/cluster/ldap`

Deletes the LDAP configuration of the cluster.

Learn more

- DOC /security/authentication/cluster/ldap

Response

```
Status: 200, Ok
```

Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Retrieve the LDAP configuration for the cluster

GET `/security/authentication/cluster/ldap`

Retrieves the cluster LDAP configuration.

Learn more

- DOC /security/authentication/cluster/ldap

Parameters

Name	Type	In	Required	Description
max_records	integer	query	False	Limit the number of records returned.
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned.
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached.
order_by	array[string]	query	False	Order results by specified fields and optional [asc
desc] direction. Default direction is 'asc' for ascending.	fields	array[string]	query	False

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
base_dn	string	Specifies the default base DN for all searches.
base_scope	string	Specifies the default search scope for LDAP queries: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
bind_dn	string	Specifies the user that binds to the LDAP servers.
bind_password	string	Specifies the bind password for the LDAP servers.
min_bind_level	string	The minimum bind authentication level. Possible values are: • anonymous - anonymous bind • simple - simple bind • sasl - Simple Authentication and Security Layer (SASL) bind
port	integer	The port used to connect to the LDAP Servers.
schema	string	The name of the schema template used by the SVM. • AD-IDMU - Active Directory Identity Management for UNIX • AD-SFU - Active Directory Services for UNIX • MS-AD-BIS - Active Directory Identity Management for UNIX • RFC-2307 - Schema based on RFC 2307 • Custom schema

Name	Type	Description
servers	array[string]	
session_security	string	Specifies the level of security to be used for LDAP communications: • none - no signing or sealing • sign - sign LDAP traffic • seal - seal and sign LDAP traffic
use_start_tls	boolean	Specifies whether or not to use Start TLS over LDAP connections.

Example response

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "base_scope": "base",
  "min_bind_level": "anonymous",
  "port": 389,
  "servers": {
  },
  "session_security": "none"
}
```

Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Update the LDAP configuration for the cluster

PATCH `/security/authentication/cluster/ldap`

Both mandatory and optional parameters of the LDAP configuration can be updated. IPv6 must be enabled if IPv6 family addresses are specified. Configuring more than one LDAP server is recommended to avoid a single point of failure. Both FQDNs and IP addresses are supported for the `servers` property. The LDAP servers are validated as part of this operation. LDAP validation fails in the following scenarios:

1. The server does not have LDAP installed.
2. The server is invalid.
3. The server is unreachable.

Learn more

- DOC /security/authentication/cluster/ldap

Parameters

Name	Type	In	Required	Description
return_records	boolean	query	False	The default is false. If set to true, the records are returned.

Request Body

Name	Type	Description
_links	_links	
base_dn	string	Specifies the default base DN for all searches.
base_scope	string	Specifies the default search scope for LDAP queries: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
bind_dn	string	Specifies the user that binds to the LDAP servers.
bind_password	string	Specifies the bind password for the LDAP servers.
min_bind_level	string	The minimum bind authentication level. Possible values are: • anonymous - anonymous bind • simple - simple bind • sasl - Simple Authentication and Security Layer (SASL) bind
port	integer	The port used to connect to the LDAP Servers.

Name	Type	Description
schema	string	The name of the schema template used by the SVM. • AD-IDMU - Active Directory Identity Management for UNIX • AD-SFU - Active Directory Services for UNIX • MS-AD-BIS - Active Directory Identity Management for UNIX • RFC-2307 - Schema based on RFC 2307 • Custom schema
servers	array[string]	
session_security	string	Specifies the level of security to be used for LDAP communications: • none - no signing or sealing • sign - sign LDAP traffic • seal - seal and sign LDAP traffic
use_start_tls	boolean	Specifies whether or not to use Start TLS over LDAP connections.

Example request

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "base_scope": "base",
  "min_bind_level": "anonymous",
  "port": 389,
  "servers": {
  },
  "session_security": "none"
}
```

Response

```
Status: 200, Ok
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
4915203	The specified LDAP schema does not exist.
4915208	The specified LDAP servers contain duplicate server entries.
4915229	DNS resolution failed due to an internal error. Contact technical support if this issue persists.
4915231	DNS resolution failed for one or more of the specified LDAP servers. Verify that a valid DNS server is configured.
23724132	DNS resolution failed for all the specified LDAP servers. Verify that a valid DNS server is configured.
4915234	Specified LDAP server is not supported because it is one of the following: multicast, loopback, 0.0.0.0, or broadcast.
4915248	LDAP servers cannot be empty or "-". Specified FQDN is not valid because it is empty or "-" or it contains either special characters or "-" at the start or end of the domain.
4915251	STARTTLS and LDAPS cannot be used together
4915257	The LDAP configuration is not valid. Verify that the Distinguished Names and bind password are correct.
4915258	The LDAP configuration is not valid. Verify that the servers are reachable and that the network configuration is correct.
23724130	Cannot use an IPv6 name server address because there are no IPv6 interfaces.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

cluster_ldap

Name	Type	Description
_links	_links	
base_dn	string	Specifies the default base DN for all searches.
base_scope	string	Specifies the default search scope for LDAP queries: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
bind_dn	string	Specifies the user that binds to the LDAP servers.
bind_password	string	Specifies the bind password for the LDAP servers.
min_bind_level	string	The minimum bind authentication level. Possible values are: • anonymous - anonymous bind • simple - simple bind • sasl - Simple Authentication and Security Layer (SASL) bind

Name	Type	Description
port	integer	The port used to connect to the LDAP Servers.
schema	string	The name of the schema template used by the SVM. • AD-IDMU - Active Directory Identity Management for UNIX • AD-SFU - Active Directory Services for UNIX • MS-AD-BIS - Active Directory Identity Management for UNIX • RFC-2307 - Schema based on RFC 2307 • Custom schema
servers	array[string]	
session_security	string	Specifies the level of security to be used for LDAP communications: • none - no signing or sealing • sign - sign LDAP traffic • seal - seal and sign LDAP traffic
use_start_tls	boolean	Specifies whether or not to use Start TLS over LDAP connections.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments

Name	Type	Description
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Create the LDAP configuration for the cluster

POST `/security/authentication/cluster/ldap`

A cluster can have only one LDAP configuration. IPv6 must be enabled if IPv6 family addresses are specified.

Required properties

- `servers` - List of LDAP servers used for this client configuration.
- `bind_dn` - Specifies the user that binds to the LDAP servers.
- `base_dn` - Specifies the default base DN for all searches.

Recommended optional properties

- `schema` - Schema template name.
- `port` - Port used to connect to the LDAP Servers.
- `min_bind_level` - Minimum bind authentication level.
- `bind_password` - Specifies the bind password for the LDAP servers.
- `base_scope` - Specifies the default search scope for LDAP queries.
- `use_start_tls` - Specifies whether or not to use Start TLS over LDAP connections.
- `session_security` - Specifies the level of security to be used for LDAP communications.

Default property values

- `schema` - *RFC-2307*
- `port` - *389*
- `min_bind_level` - *simple*
- `base_scope` - *subtree*
- `use_start_tls` - *false*
- `session_security` - *none*

Configuring more than one LDAP server is recommended to avoid a single point of failure. Both FQDNs and IP addresses are supported for the `servers` property. The LDAP servers are validated as part of this operation.

LDAP validation fails in the following scenarios:

1. The server does not have LDAP installed.
2. The server is invalid.
3. The server is unreachable.

Learn more

- DOC /security/authentication/cluster/ldap

Parameters

Name	Type	In	Required	Description
return_records	boolean	query	False	The default is false. If set to true, the records are returned.

Request Body

Name	Type	Description
_links	_links	
base_dn	string	Specifies the default base DN for all searches.
base_scope	string	Specifies the default search scope for LDAP queries: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
bind_dn	string	Specifies the user that binds to the LDAP servers.
bind_password	string	Specifies the bind password for the LDAP servers.

Name	Type	Description
min_bind_level	string	The minimum bind authentication level. Possible values are: • anonymous - anonymous bind • simple - simple bind • sasl - Simple Authentication and Security Layer (SASL) bind
port	integer	The port used to connect to the LDAP Servers.
schema	string	The name of the schema template used by the SVM. • AD-IDMU - Active Directory Identity Management for UNIX • AD-SFU - Active Directory Services for UNIX • MS-AD-BIS - Active Directory Identity Management for UNIX • RFC-2307 - Schema based on RFC 2307 • Custom schema
servers	array[string]	
session_security	string	Specifies the level of security to be used for LDAP communications: • none - no signing or sealing • sign - sign LDAP traffic • seal - seal and sign LDAP traffic
use_start_tls	boolean	Specifies whether or not to use Start TLS over LDAP connections.

Example request

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "base_scope": "base",
  "min_bind_level": "anonymous",
  "port": 389,
  "servers": {
  },
  "session_security": "none"
}
```

Response

```
Status: 201, Created
```

Name	Type	Description
_links	_links	
num_records	integer	Number of LDAP records.
records	array[ldap_service]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "base_scope": "base",
    "min_bind_level": "anonymous",
    "port": 389,
    "preferred_ad_servers": {
    },
    "servers": {
    },
    "session_security": "none",
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    }
  }
}
```


Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
4915203	The specified LDAP schema does not exist.
4915207	The specified LDAP servers contain duplicate server entries.
4915229	DNS resolution failed due to an internal error. Contact technical support if this issue persists.
4915231	DNS resolution failed for one or more of the specified LDAP servers. Verify that a valid DNS server is configured.
23724132	DNS resolution failed for all the specified LDAP servers. Verify that a valid DNS server is configured.
4915234	The specified LDAP server is not supported because it is one of the following: multicast, loopback, 0.0.0.0, or broadcast.
4915248	LDAP servers cannot be empty or "-". Specified FQDN is invalid because it is empty or "-" or it contains either special characters or "-" at the start or end of the domain.
4915251	STARTTLS and LDAPS cannot be used together.
4915257	The LDAP configuration is invalid. Verify that bind-dn and bind password are correct.
4915258	The LDAP configuration is invalid. Verify that the servers are reachable and that the network configuration is correct.
23724130	Cannot use an IPv6 name server address because there are no IPv6 interfaces.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

cluster_ldap

Name	Type	Description
_links	_links	
base_dn	string	Specifies the default base DN for all searches.
base_scope	string	Specifies the default search scope for LDAP queries: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
bind_dn	string	Specifies the user that binds to the LDAP servers.
bind_password	string	Specifies the bind password for the LDAP servers.
min_bind_level	string	The minimum bind authentication level. Possible values are: • anonymous - anonymous bind • simple - simple bind • sasl - Simple Authentication and Security Layer (SASL) bind

Name	Type	Description
port	integer	The port used to connect to the LDAP Servers.
schema	string	The name of the schema template used by the SVM. • AD-IDMU - Active Directory Identity Management for UNIX • AD-SFU - Active Directory Services for UNIX • MS-AD-BIS - Active Directory Identity Management for UNIX • RFC-2307 - Schema based on RFC 2307 • Custom schema
servers	array[string]	
session_security	string	Specifies the level of security to be used for LDAP communications: • none - no signing or sealing • sign - sign LDAP traffic • seal - seal and sign LDAP traffic
use_start_tls	boolean	Specifies whether or not to use Start TLS over LDAP connections.

_links

Name	Type	Description
next	href	
self	href	

svm

Name	Type	Description
_links	_links	
name	string	The name of the SVM.

Name	Type	Description
uuid	string	The unique identifier of the SVM.

ldap_service

Name	Type	Description
_links	_links	
ad_domain	string	This parameter specifies the name of the Active Directory domain used to discover LDAP servers for use by this client. This is mutually exclusive with `servers` during POST and PATCH.
base_dn	string	Specifies the default base DN for all searches.
base_scope	string	Specifies the default search scope for LDAP queries: • base - search the named entry only • onelevel - search all entries immediately below the DN • subtree - search the named DN entry and the entire subtree below the DN
bind_dn	string	Specifies the user that binds to the LDAP servers.
bind_password	string	Specifies the bind password for the LDAP servers.
min_bind_level	string	The minimum bind authentication level. Possible values are: • anonymous - anonymous bind • simple - simple bind • sasl - Simple Authentication and Security Layer (SASL) bind

Name	Type	Description
port	integer	The port used to connect to the LDAP Servers.
preferred_ad_servers	array[string]	
schema	string	The name of the schema template used by the SVM. • AD-IDMU - Active Directory Identity Management for UNIX • AD-SFU - Active Directory Services for UNIX • MS-AD-BIS - Active Directory Identity Management for UNIX • RFC-2307 - Schema based on RFC 2307 • Custom schema
servers	array[string]	
session_security	string	Specifies the level of security to be used for LDAP communications: • none - no signing or sealing • sign - sign LDAP traffic • seal - seal and sign LDAP traffic
svm	svm	
use_start_tls	boolean	Specifies whether or not to use Start TLS over LDAP connections.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

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